Arista Access Router (MAR-ACC-04) - Routing Protocol Flap (BGP, OSPF, ISIS) SOP

Node: MAR-ACC-04 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_8389

Description

This Standard Operating Procedure (SOP) addresses routing protocol flap failures on the Arista Access Router (MAR-ACC-04) at the Marathahalli Core site, specifically related to BGP route churn, OSPF adjacency loss, and ISIS LSP flooding. The underlying cause of this failure domain is attributed to the dynamic nature of routing protocols, particularly in environments with high network volatility, where changes in the routing topology can trigger frequent routing updates, leading to protocol flapping. In the context of Arista equipment, this issue may be exacerbated by factors such as misconfigured protocol timers, incorrect router ID settings, or faulty hardware components. Furthermore, the role of the Access Router in this scenario is critical, as it serves as the entry point for customer traffic, making it essential to promptly diagnose and resolve any routing protocol flap issues to maintain network stability and ensure uninterrupted service delivery.

Pre-requisite Checks

- Pre-requisite: Verify that the Marathahalli Core site is accessible and that remote hands are available for assistance, if required.

- Pre-requisite: Ensure that the redundant link status is checked to confirm that the network has adequate redundancy to handle potential failures.

Potential Root Causes

Cause	Probability
Misconfigured protocol timers	High
Incorrect router ID settings	Medium
Faulty hardware components	Low

Diagnostic Commands & Expected Outputs

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Command: show bgp summary, Expected Output: BGP router identifier 192.168.1.1, local AS number 65001 Status codes: s-suppressed, d-dampened, h-hold, *-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i - IGP, e - EGPs, ? - incomplete Network Next Hop Weight Path *> 10.1.1.0/24 10.1.1.2 0 65002 65001 i, Description: This command is used to display a summary of the BGP routing table, including the router ID, local AS number, and the status of each network.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Interface FF DBY Flood Queue 192.168.1.2 1 FULL/ - 00:00:32 Ethernet1/1/1 0/0/0 , Description: This command displays information
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about OSPF neighbors, including the neighbor ID, priority, state, and interface.

Command: show isis database, Expected Output: IS-IS Level-1 Link-State Database: L1/L2 Protocol L1/L2 Metric L1/L2 TID Sequence Checksum Age Interface L1 0x0000000B 0x00000020 0x00000000 0x03A7F3A8 0x00000005 0x00000000 Ethernet1/1/1 , Description: This command displays the ISIS database, including the L1/L2 protocol information, L1/L2 metric, and sequence number.
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Resolution Steps

Step 1: Step: Verify that the protocol timers are configured correctly, Command: show run | include timer, Description: Check the configuration for the protocol timers to ensure they are set to the recommended values.

Step 2: Step: Check the router ID settings, Command: show run | include router-id, Description: Verify that the router ID is set correctly and is unique within the network.

Step 3: Step: Disable the faulty hardware component, Command: interface Ethernet1/1/1 shutdown , Description: Disable the faulty hardware component to prevent further issues.

Post-Check Verification

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Command: show bgp summary, Expected Output: BGP router identifier 192.168.1.1, local AS number 65001 Status codes: s-suppressed, d-dampened, h-hold, *-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i - IGP, e - EGPs, ? - incomplete Network Next Hop Weight Path *> 10.1.1.0/24 10.1.1.2 0 65002 65001 i, Description: Verify that the BGP routing table is stable and that the protocol flap has been resolved.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Interface FF DBY Flood Queue 192.168.1.2 1 FULL/ - 00:00:32 Ethernet1/1/1 0/0/0 , Description: Verify that the OSPF neighbor relationships are stable and that the protocol flap has been resolved.

Command: show isis database, Expected Output: IS-IS Level-1 Link-State Database: L1/L2 Protocol L1/L2 Metric L1/L2 TID Sequence Checksum Age Interface L1 0x0000000B 0x00000020 0x00000000 0x03A7F3A8 0x00000005 0x00000000 Ethernet1/1/1 , Description: Verify that the ISIS database is stable and that the protocol flap has been resolved.
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Roll-back Procedures

- Step: Restore the original configuration, Command: copy run start, Description: Restore the original configuration to ensure the network returns to its pre-change state.

- Step: Re-enable the faulty hardware component, Command: interface Ethernet1/1/1 no shutdown , Description: Re-enable the faulty hardware component once it has been debugged.